# Samaranjan Manjhi

**C/C++ Systems & Cyber Security Software Developer**

+91 9082650120 | samaranjanedu01@gmail.com | LinkedIn | Portfolio (Video Demos) | Mumbai, India

## SUMMARY

C/C++ Systems and Cyber Security Engineer with **3+ years** of experience building production-grade endpoint security tools, kernel monitors, binary integrity frameworks, and Data Loss Prevention (DLP) systems for **macOS and Linux**. Deep expertise in Endpoint Security Framework (ESF), execution gating, system call hooking, Qt GUI development, and cross-platform C++ architecture. Delivered **10+ enterprise security modules** deployed globally across macOS and Linux endpoints for enterprise antivirus suites.

## TECHNICAL SKILLS

**Languages:** C, C++ (Modern C++), Bash Scripting, Relational SQL
**Systems & Architecture:** Multithreading, Inter-Process Communication (IPC), OS Internals (Linux/macOS), Memory Management, TCP/IP Networking, Data Structures
**Frameworks & Kernel:** Endpoint Security Framework (ESF), Extended Berkeley Packet Filter (eBPF), System Audit, Core Foundation
**Tools & Environment:** Git, CMake, Make, Xcode, GDB, Valgrind, KDB/KGDB, SQLite, Boost, OpenSSL

## PROFESSIONAL EXPERIENCE

**Systems & Security Software Engineer** | **MicroWorld Software Services Pvt. Ltd. — Mumbai** *Aug 2023 – Present*

- Designed and delivered **10+ production security modules** spanning kernel monitors, DLP engines, binary integrity systems, firewall rule generators, and cloud backup agents — deployed globally across enterprise Linux and macOS systems running security products.
- Constructed high-performance, cross-platform C/C++ daemons utilizing POSIX Unix domain socket IPC, reducing system memory overhead by 25% and increasing event detection throughput by 40%.
- Independently owned end-to-end design, implementation, and deployment for select modules, coordinating with QA and product teams to define requirements and validate detection accuracy prior to release.
- Regularly mentored and onboarded new engineers on secure C/C++ development, kernel-level debugging, and internal security frameworks, shortening ramp-up time on kernel and DLP module development.

## KEY PROJECTS

**Real-Time Virus Scanning — eBPF** | **C/C++, eBPF, Kernel Tracing, AV Engine Integration** *Mar 2026 – Jun 2026*

- Engineered a lock-free audit map and a dedicated C++ thread pool to optimize event throughput, sustaining real-time detection under heavy file I/O operations without blocking critical system processes.
- Integrated quarantine and process-termination actions triggered on malware detection, with a configurable watchlist and root/parent-process validation to reduce false positives and redundant re-scans.
- Optimized event throughput with a lock-free audit map and dedicated scanner thread pool, sustaining real-time on-access detection under heavy file I/O without blocking user processes.

**MW Guardian — Binary Integrity & Execution Gating** | **C/C++, Linux Kernel, Cryptographic Verification** *Feb 2026 – Jul 2026*

- Enforced execution gating via **kernel permission events** to intercept binary execution at kernel level, synchronously validating cryptographic signatures and terminating untrusted binaries prior to execution.
- Architected a multi-key cryptographic signing pipeline supporting key rotation while retaining legacy public keys to preserve signature validity for previously signed binaries.
- Created startup self-verification routines where guardian and verifier components validate their own binary signatures prior to initiating the main event loop, eliminating bootstrap trust gaps.

**Two-Factor Authentication for SSH** | **C/C++, PAM Module, TOTP Cryptography, Qt Framework, macOS** *Jul 2026 – Aug 2026*

- Extended the PAM authentication stack with a custom TOTP verification module, layering a second authentication factor onto SSH login while preserving compatibility with existing PAM configuration and login flows.
- Built and integrated a Qt-based 2FA enrollment and management widget directly into the production macOS security console GUI, allowing administrators to enable, configure, and audit 2FA policy per endpoint.
- Implemented secret provisioning and QR-code based device enrollment, coordinating configuration state between the PAM module and GUI layer without requiring a service restart.

**Printer Control & Data Loss Prevention (DLP)** | **macOS, C/C++, ESF, Print Service Libraries, Xcode** *Oct 2025 – Jan 2026*

- Developed a macOS DLP driver utilizing ESF and print spool monitoring to inspect print files for sensitive information, cancelling unauthorized print jobs prior to printing.
- Applied application-level process whitelisting and blacklisting enforced via ESF process identity, bypassing clean content for trusted applications and blocking unauthorized applications outright.
- Archived scanned print jobs alongside result metadata to maintain audit trails for compliance reporting and post-event forensic investigations.

**File Integrity Monitoring (FIM)** | **macOS, C/C++, File System Events, SQLite Database, CommonCrypto MD5** *Jun 2025 – Aug 2025*

- Built a file system event daemon storing cryptographic file hashes, permissions, file sizes, and timestamps in SQLite, emitting real-time security alerts upon unauthorized path modifications, renames, or deletions.
- Established baseline state snapshots on initial startup and executed diff comparisons on subsequent runs to output structured security event logs across configurable watch paths.

**Content Scanner — DLP Engine & CLI** | **C/C++, Regex Engine, Optical Character Recognition, Archive Libraries** *Sep 2024 – May 2025*

- Built a multi-threaded DLP scanner supporting **15+ file formats** via regular expression parsing and optical character recognition, packaged as a shared library and CLI utility.
- Architected a high-performance Unix domain socket server in C++ allowing asynchronous client streaming; eliminated redundant regex engine initializations to slash per-scan latency by 60%.

## EDUCATION

**Bachelor of Engineering — Computer Engineering** *2019 – 2023*

Shree L.R. Tiwari College of Engineering, Mumbai University — CGPA: **8.56 / 10**